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PHYTOREMEDIATION FOR CO-CONTAMINATED SOILS OF CADMIUM AND POLYCHLORINATED BIPHENYLS USING ORNAMENTAL PLANT *TAGETES PATULA* (size 14)

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ABSTRACT: (~250 words)

In this study, pot-culture experiments conducted to investigate the single effect of Cd, PCBs and the joint effect of Cd-PCBs with *Tagetes patula.* The study highlights that the minimum concentration of PCBs (100 µg kg^{-1}) could enable the growth of the plant in that the biomass increased by 27.76% when compared to control. In all the experiments done so far, the Cd concentrations over the surface parts were found to be above 100 mg kg^{-1}. Significant positive correlations were observed between the Cd and PCBs concentrations accumulated in tissues of the soil and plants ($P<0.05$). The *T. patula* exhibited high tolerance to Cd and PCBs, and the plant promoted the removal rate of PCBs. The removal rates of PCB18 and PCB28 were up to 42.72 and 42.29%, respectively. The study highlights the potential and suitability of *T. patula* for phytoremediation of Cd and PCBs in contaminated soils.

Keywords: Phytoremediation; *T. patula;* Contaminated soils; Pot-culture; Cd, PCBs